

Supplementary material

How much does the map of science depend on the embedding model?

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This supplement supplies definitions, checks and complete result files. Each table has a printed summary and unrounded CSV files in `tables/data/Sxx/`. Figure data are in `figures/data/`. Summaries do not replace those files.

Which articles belong to which check? The 500,000 records are the starting collection. The text experiment uses 52,000 of them to compare title, abstract and both; its earlier 26,000-paper trial is included, not added. The geometric analysis in S6–S7 reuses that same set of 52,000. A different set of 52,000 is used for the shared-passage check in S1. The matched searches in S4 follow 10,850 starting papers while changing the papers available around them. These are overlapping uses of one collection, not separate downloads.

Here, Field means one of the 26 broad research areas and Subfield means a narrower specialty. The main article introduces the comparisons in that order; this supplement keeps the database names and technical definitions so that the calculations can be checked.

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S1 Corpus, models and provenance

S1.1 Eligibility, quotas and quality flags

OpenAlex’s core catalogue was queried for 2000–2024 records of type article, review or conference-paper, with an English language label, an abstract, no retraction or paratext flag and a known primary Field. Local preparation required a non-empty title and at least 50 regex words in the reconstructed abstract. It preserved literal content rather than translating or generating missing summaries. The unit was a distinct OpenAlex work ID, not a deduplicated intellectual work.

Seeded random request blocks supplied the general cohort. Saved candidates were processed in their recorded order until 400,000 eligible, distinct IDs were retained. The 100,000-record supplement was assigned to the smallest of the 130 Field-period groups, with deterministic Field/year tie-breaking. This raised minimum coverage without reducing the larger groups. Exact year counts were retained rather than forcing equal quotas within each five-year period. Raw retrieval ended on 15 September 2026; API responses were collected over time, so the corpus is not a simultaneous database snapshot. Saved responses, ID order and content hashes are required to reconstruct it; seeds alone would not reproduce a changing API.

Cleaning rejected recognised non-abstract content, explicit withdrawal/retraction titles and clear language failures. For language, fastText’s lid.176 model was applied to the whole abstract and three consecutive parts. A record was excluded only if the whole text had a non-English top label with score at least 0.9 and all three parts agreed on that language with scores at least 0.8. These are conservative operational scores, not calibrated probabilities or independent tests. Remaining uncertainty, possible mixed language, 50–79-word or unusually long abstracts, possible notices and duplicate DOI/text were marked. Content rules were developed by inspecting the downloaded corpus before embedding; they were not preregistered before extraction.

The final corpus retains 9,883 abstracts with language uncertainty, including 3,002 flagged as potentially mixed. These sets overlap. It also includes 42,849 abstracts of 50–79 words and 249 over 2,000 words. Subsequent analysis uses the flags for sensitivity rather than silently changing the corpus. Source checks establish agreement with downloaded records, not that every record is correct. Table S1 supplies the distribution and the full flag tables.

Table S1: Clean corpus by Field, period and cohort.

Field	2000-04	2005-09	2010-14	2015-19	2020-24	Base	Extra
Agricultural and Biological Sciences	2,166	2,528	3,940	4,298	6,491	18,920	503
Arts and Humanities	2,166	2,166	2,186	2,732	3,380	10,696	1,934
Biochemistry, Genetics and Molecular Biology	2,739	4,071	5,251	5,596	6,840	24,497	0
Business, Management and Accounting	2,166	2,166	2,186	2,660	4,040	11,137	2,081
Chemical Engineering	2,166	2,166	2,166	2,166	2,166	1,195	9,635
Chemistry	2,166	2,166	2,166	2,166	2,166	7,290	3,540
Computer Science	3,310	5,594	7,094	6,858	9,972	32,828	0
Decision Sciences	2,166	2,166	2,166	2,166	2,166	3,817	7,013
Earth and Planetary Sciences	2,166	2,166	2,166	2,166	2,166	7,256	3,574
Economics, Econometrics and Finance	2,166	2,166	2,166	2,166	2,845	8,620	2,889
Energy	2,166	2,166	2,166	2,166	2,166	2,617	8,213
Engineering	6,990	11,310	15,203	14,738	18,907	67,148	0
Environmental Science	2,166	2,876	4,005	4,018	5,919	18,569	415
Immunology and Microbiology	2,166	2,166	2,166	2,166	2,166	4,456	6,374
Materials Science	2,166	2,166	2,962	3,152	3,787	13,335	898
Mathematics	2,166	2,166	2,166	2,166	2,166	4,581	6,249
Medicine	7,128	10,946	15,771	18,629	26,195	78,669	0
Neuroscience	2,166	2,166	2,166	2,166	2,166	6,824	4,006
Nursing	2,166	2,166	2,166	2,166	2,166	2,077	8,753
Pharmacology, Toxicology and Pharmaceutics	2,166	2,166	2,166	2,166	2,166	1,715	9,115
Physics and Astronomy	2,166	2,704	3,094	3,388	3,470	14,772	50
Psychology	2,166	2,166	2,166	2,577	3,970	10,816	2,229
Social Sciences	2,776	4,893	7,110	8,891	13,879	37,549	0
Veterinary	2,166	2,166	2,165	2,165	2,165	642	10,185
Dentistry	2,165	2,165	2,165	2,165	2,165	2,238	8,587
Health Professions	2,165	2,165	2,165	2,165	2,833	7,736	3,757
Total	68,427	83,908	101,289	107,858	138,518	400,000	100,000

Total: 500,000 distinct OpenAlex work IDs. Period columns contain base plus extra; the last two columns divide those same articles and must not be added again. Core catalogue; years 2000–2024; article, review and conference-paper; OpenAlex English label, non-empty title, reconstructible abstract with at least 50 regex words, known primary Field, no retraction or paratext flag. Additional cleaning rejects clear language/content failures and marks ambiguous cases. No claim that every retained abstract is exclusively English or correctly labelled. Exact annual/cohort counts and quality flags are included as CSVs; raw retrieval ended on 15 September 2026. The database was not a simultaneous snapshot.

S1.2 Checkpoints, input and truncation

All ten models use the same ordered IDs and original title/abstract text. The customary separators, full model revisions and any adapter revisions are listed in Table S2. Outputs were saved as 32-bit vectors; subsequent geometric calculations use 64-bit arithmetic. Missing or degenerate outputs were checked, and each run records code, configuration, file and environment hashes.

Mean pooling averages every non-padding last-layer position, including model-added special tokens. CLS is the first-position state before any pooler; SEP is the final non-padding position, rather than the separator between title and abstract. Only BERT, SciBERT, BioBERT and PubMedBERT change under the CLS/SEP scenarios. SimCSE uses its pre-pooler CLS state; SPECTER2 uses its proximity adapter. MPNet and MiniLM retain their sentence-transformer mean-pooling configuration. Readout scenarios are not additional independently trained models.

The common-fragment comparison finds a shared prefix of content characters that fits every model, keeping each tokenizer and its customary separators. It uses a separate 52,000-ID panel and compares that fragment with the native representation of those same IDs. It does not equate token sequences between vocabularies. MiniLM’s separate 512-token check uses its supported architectural limit; vectors for texts unaffected by the original truncation are reused exactly. Neither check demonstrates that longer input gives better thematic representations.

Table S2: Frozen checkpoints, text formats and native truncation.

Model	Checkpoint and full revision	Input / truncation
SPECTER	allenai/specter	Title [SEP] abstract
	Revision: 81cbfb43d4fc2e728d5b4201ce14987db8d	Limit: 512
	0854c	Truncated: 5.66%
SPECTER2	allenai/specter2_base	Title [SEP] abstract
	Revision: 3447645e1def9117997203454fa4495937b	Limit: 512
	fbd83	Truncated: 5.00%
	Adapter: allenai/specter2	
SciNCL	2081559630a80fc5851d8f798a05ba81e9468089	
	malteos/scincl	Title [SEP] abstract
	Revision: ebc5348d184ba2fc9beee69b4e394263fce	Limit: 512
SciBERT	57b2e	Truncated: 5.00%
	allenai/scibert_scivocab_uncased	Title, blank line,
	Revision: 24f92d32b1bfb0bc9ab193ff3ad01e877	abstract
BERT	32fc1	Limit: 512
		Truncated: 4.96%
	google-bert/bert-base-uncased	Title, blank line,
	Revision: 86b5e0934494bd15c9632b12f734a8a67f7	abstract
MPNet	23594	Limit: 512
		Truncated: 6.87%
	sentence-transformers/all-mpnet-base-v2	Title, blank line,
MiniLM	Revision: e8c3b32edf5434bc2275fc9bab85f82640a	abstract
	19130	Limit: 384
		Truncated: 20.16%
	sentence-transformers/all-MiniLM-L6-v2	Title, blank line,
PubMedBERT	Revision: 1110a243fdf4706b3f48f1d95db1a4f5529	abstract
	b4d41	Limit: 256
		Truncated: 49.58%
	microsoft/BiomedNLP-BiomedBERT-base-uncased	Title, blank line,
BioBERT	-abstract-fulltext	abstract
	Revision: e1354b7a3a09615f6aba48dfad4b7a613ee	Limit: 512
	f7062	Truncated: 4.75%
SimCSE	dmis-lab/biobert-v1.1	Title, blank line,
	Revision: 551ca18efd7f052c8dfa0b01c94c2a8e68b	abstract
	c5488	Limit: 512
SimCSE		Truncated: 9.00%
	princeton-nlp/unsup-simcse-bert-base-uncas	Title, blank line,
	ed	abstract
SimCSE	Revision: 218baf6bc48c463a53f5b4e1c568299537f	Limit: 512
	ab124	Truncated: 6.87%

Truncation is measured on the full 500,000-article title-plus-abstract corpus and is not an error rate. The principal MiniLM limit remains 256 tokens; its 512-token variant is a separate sensitivity. CLS/SEP alternatives apply only to BERT, SciBERT, BioBERT and PubMedBERT. SEP takes the last non-padding position, not the title separator. Tokenizer differences mean that equal token limits would not imply equal text. Model-card URLs and all revisions are in the CSV.

S2 Agreement measures and sampling stability

This section explains how agreement between two models is calculated, then checks how much the result changes when other articles are selected. Those are two different questions.

S2.1 Coverage and chronology

Table S3: Coverage and chronology of the comparison blocks.

Block	Coverage	When specified	Interpretation
Initial comparison	500,000 corpus; 130 cells; 45 fixed model pairs	Before this phase’s agreements	Normalised vectors; corrected CKA, k25; CLS/SEP and size checks.
Common fragment	52,000 paired IDs, distinct from the input panel	Before first scientific comparison	Same content characters; native tokenizers and separators.
Centres / random groups	256 per centre; 26/217/26 centres	Added after initial results	Matched dates/sizes; addresses centre averaging, not thematic validity.
Input expansion	26,000 to 52,000 nested IDs	After pilot; before expanded result	Three inputs and pooling crossed; 100 comparable selections.
Matched local searches	217 groups, 50 queries, ten selections	Expanded phase design	256 candidates per scope, including identical queries and matched dates.
Atlas	217 Subfields; 183 in sensitivity panel; 12 paper examples	After global results; before reading selected titles	Rule-selected examples; source errors retained.
Morphology	52,000 IDs; alternative measures and synthetic clouds	Exploratory extension after core results	Protocol before new real-data metrics. Neighbour-fraction check added during execution.
Field pairs	325 pairs per property; 26 saved conditions	After morphology results	Direction/magnitude rules fixed before new pair counts; no new inference.

This is a sequence of documented phases, not a preregistered study. Sources: ANALYSIS_PROTOCOL, ROBUSTNESS_PROTOCOL, CASE_ATLAS_PROTOCOL, MORPHOLOGY_PROTOCOL and FIELD_PAIR_PROTOCOL, with their dated decision entries. Later controls are not relabelled as initial hypotheses.

S2.2 Alignment and neighbour definitions

For each comparison, rows of X and Y identify the same articles. Vectors are scaled to unit length, and columns are then centred for CKA and Procrustes. Let n be the row count and define

$$\begin{aligned} S_{XY} &= \|X^\top Y\|_F^2, & R_{XY} &= \sum_i \|X_i\|^2 \|Y_i\|^2, \\ T_X &= \sum_i \|X_i\|^2, & H_{XY} &= S_{XY} - \frac{n}{n-2} R_{XY} + \frac{T_X T_Y}{(n-1)(n-2)}. \end{aligned}$$

The reported corrected alignment is

$$\text{CKA}_{\text{corr}}(X, Y) = \frac{H_{XY}}{\sqrt{H_{XX} H_{YY}}}.$$

The common HSIC factor cancels. Negative corrected values are retained, and degenerate inputs are rejected. The implementation was checked independently using relation matrices with zero diagonals. The correction addresses finite-sample inflation (Kornblith et al., 2019; Murphy et al., 2024); the complete ratio is not asserted to be exactly unbiased.

Procrustes similarity is $\|X^\top Y\|_*/\sqrt{T_X T_Y}$, where the numerator is the sum of singular values. Its arccosine gives the reported angular distance. Rotation/reflection and a global scale can be aligned; an arbitrary deformation cannot. Unequal embedding dimensions can be understood by adding zero coordinates to the smaller representation. Normalisation precedes centring, so this procedure does not imply invariance to translating the original vectors and normalising them again (Williams et al., 2021).

The distance-rank check correlates cosine distances for 20,000 distinct article pairs selected with a recorded seed, identically in both representations. Ties receive average ranks. These pair observations and the alternative alignment measures are not treated as independent replications (Williams, 2024).

For query i and models a, b , neighbour overlap is

$$O_k(i; a, b) = \frac{|N_k^a(i) \cap N_k^b(i)|}{k}.$$

The query is excluded by ID. In the core exact search, similarities are computed in 64-bit blocks and rounded to twelve decimal places for deterministic tie resolution by global

row order. Independent sorting checks verify the stored lists. The combinatorial reference for independent uniform lists among M eligible candidates is k/M ; the adjusted overlap is $(O_k - k/M)/(1 - k/M)$. This reference does not remove density, popular-neighbour or composition effects. The primary results therefore retain raw overlap alongside explicitly matched candidate pools.

S2.3 Selection screens and retained alerts

The original screen uses twenty selections at each of 1,024 and 2,048 articles per Field-period group, paired between models. It flags an absolute median shift above 0.02 or a central 95% selection width above 0.04 at the larger size. A row permutation per model/group provides an alignment diagnostic rather than a permutation-test p -value. The quality sensitivity retains one representative per connected duplicate DOI/text component and excludes specified flags, leaving 448,886 records. It changes the operational population and does not replace the main corpus.

Subfield screens use twenty nested selections up to 256/512 articles, subject to availability. Groups below eight articles lack a two-size diagnosis. If the larger draw equals the full group, its zero selection width is explicitly marked and cannot establish precision for a wider population. Table S4 keeps these cases separate from the original Field-period alerts.

Table S4: Agreement of alternative measures and retained stability alerts.

Measures compared	Cells	Mean rank agreement	Minimum	Maximum
Corrected CKA / Procrustes	130	0.891	0.793	0.960
Corrected CKA / Distance ranks	130	0.881	0.777	0.963
Procrustes / Distance ranks	130	0.789	0.649	0.893

The top panel reports descriptive Spearman correlations of model-pair agreements within each of 130 Field-period cells. These correlations compare measures; they do not validate thematic content. All 41 saved summaries and all individual original alerts accompany the table.

- Original Field-period screen: 50 alerts among 5,850 comparisons; largest central-95% selection width among flagged cases: 0.064.
- The larger sample does not equal the entire group: 2,628 alerts / 8,235 comparisons; 183 Subfields.
- The larger sample equals the entire group: 213 alerts / 3,060 comparisons; 68 Subfields.

A zero selection width when the whole group is used does not establish stability at a smaller size. These are within-corpus screens, not population confidence intervals. Alerts from different designs remain separate.

S3 Text input, pooling and truncation

This experiment keeps the papers fixed and changes what each model reads. The main question is whether that change matters more or less than choosing another model.

S3.1 Model-specific input effects

The title/abstract experiment uses 52,000 articles: 400 per Field-period group and 2,000 per Field, selected by a fixed ID-based order. The first trial used 26,000; all those IDs remain in the larger set. For each model, the model-change effect is one minus mean agreement with the other nine models on title plus abstract. The input-change effect is one minus agreement between that model’s combined input and its title-only or abstract-only input. Their difference is positive when model change is larger. Neighbour changes use the same 2,000 candidates per Field; self is excluded.

A symmetric summary of input transitions checks dependence on the combined-input reference. Mean, CLS and SEP scenarios cross all three inputs; only the four word-level BERTs change readout, which also changes the cross-model reference. Saved combined-input and pilot vectors were reused with exact ID/text alignment and bitwise checks. The analysis makes a practical comparison of input conditions, not a causal decomposition of architecture and content.

Table S5: Input changes by model, averaged equally over the 26 Fields.

Model	Title: shape	Abstract: shape	Title: neighbours	Abstract: neighbours
SPECTER	0.208	0.044	61.1%	31.3%
SPECTER2	0.203	0.048	60.6%	31.3%
SciNCL	0.205	0.028	57.3%	23.9%
SciBERT	0.532	0.014	84.2%	15.8%
BERT	0.418	0.013	83.6%	18.0%
MPNet	0.189	0.022	51.7%	17.7%
MiniLM	0.246	0.031	58.1%	21.5%
PubMedBERT	0.526	0.017	82.9%	16.2%
BioBERT	0.636	0.024	83.4%	19.2%
SimCSE	0.496	0.028	83.8%	26.3%

Same 52,000 IDs and 2,000 candidates per Field. Shape change: 1 minus corrected CKA. Neighbour change: fraction of the 25 neighbours replaced. Title/abstract conditions compare each model with its own full input. Model-choice means and all Field-specific values are included as CSVs, alongside the full recipe cross. These are practical input changes with native text limits, not a causal separation of encoder and text length.

Title-only model-minus-input contrast (positive: changing model has the larger mean effect): MEAN, shape = 0.0042; MEAN, neighbours = -0.0191; CLS, shape = 0.0363; CLS, neighbours = 0.0304; SEP, shape = 0.0627; SEP, neighbours = 0.0171. Only the four word BERTs change readout in CLS/SEP. Close means do not establish equivalence.

S3.2 The 26k to 52k stability check

Both sizes were assessed with 20 and then 100 selections without replacement, using 500 articles per Field in the pilot and 1,000 in the expansion, balanced across dates. The first twenty repeats reproduce the earlier diagnostic. The same 0.02 median-shift and 0.04 central-width rules apply to the principal shape contrast. Retaining the original 31 keys distinguishes persistent, resolved and newly appearing alerts. The 100-selection comparison yields 72 to five alerts, while the historical twenty-selection comparison yields 31 to four. These counts must not be substituted for one another. They do not extend automatically to every readout or neighbour analysis.

Table S6: Input-alert transitions for the nested 26k and 52k panels.

Selections / scope	Contrasts	26k alerts	52k alerts	Persist	Resolved	New
20 / model-Field	520	31	4	3	28	1
20 / Field average	52	0	0	0	0	0
100 / model-Field	520	72	5	5	67	0
100 / Field average	52	0	0	0	0	0

The five remaining model-Field-input alerts with 100 selections are: BioBERT / Agricultural and Biological Sciences / abstract (width 0.047); BioBERT / Earth and Planetary Sciences / abstract (width 0.045); BioBERT / Economics, Econometrics and Finance / abstract (width 0.042); BioBERT / Immunology and Microbiology / abstract (width 0.063); PubMedBERT / Health Professions / title (width 0.040). No alert does not imply an effect different from zero. Original 20-selection and later 100-selection counts are kept separate.

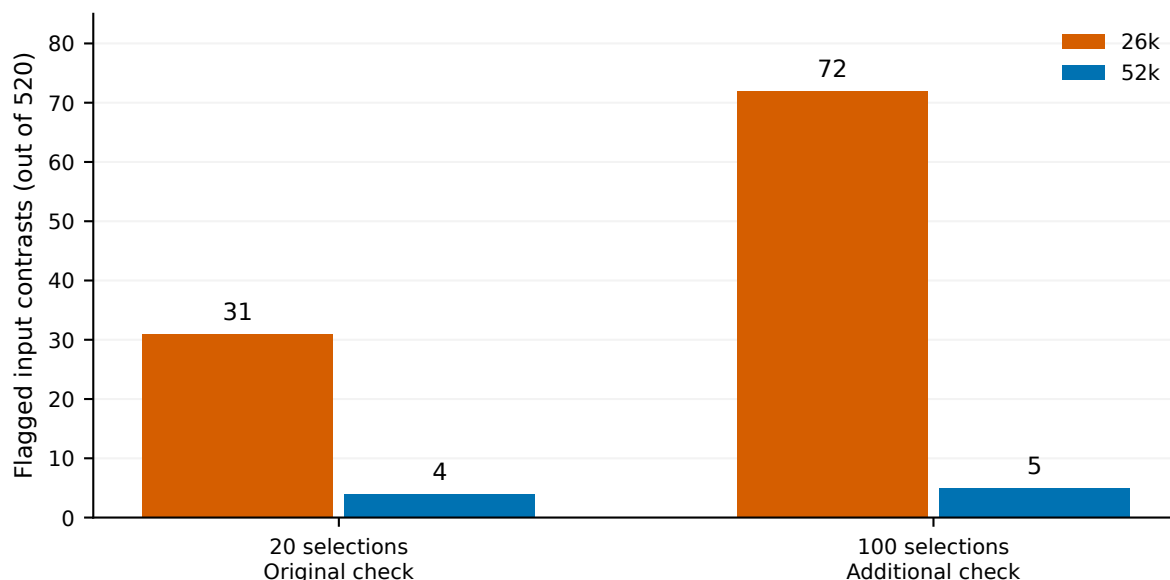


Figure S1: Input stability alerts for nested samples. Counts are model–Field–input contrasts out of 520. The original 20-selection comparison and later 100-selection check use the same recorded thresholds, but their counts are not interchangeable. These are selection-sensitivity checks within the fixed corpus. The 52 Field-averaged contrasts are reported separately in Table S6.

S4 Scale, neighbourhoods and concrete cases

These checks move from broad areas to specialties and individual papers. They separate a change in search scope from a change in the number of available papers.

S4.1 Matched scopes and candidate sizes

Within-group comparisons use 128, 256 and 512 articles, with nested ID-based selections and the same 183 eligible Subfields at all three sizes. The native analysis retains all 252 Subfields where the measure is defined; groups with exactly $k + 1$ articles have uninformative complete neighbour lists and are marked accordingly. A separate time comparison retains the same 125 Subfields with at least 128 articles in every period.

Centre comparisons average unit article vectors and scale each centre to unit length before corrected CKA. Direct and weighted-Subfield reconstructions of Field centres were checked for equality. The original matched centres contain 256 articles per group. Its twenty restricted selections change which Subfield represents each Field, not the articles within those Subfields. The repeated centre, size, matched-random and omission analyses now reported

in the main text are specified in S9. Random references retain cross-model article correspondence; they are not null tests of independence.

The paired search fixes 50 queries per eligible Subfield, with 256 candidates in each scope and ten selections. The same 50 queries are included in both candidate sets; only the other 206 positions change scope. Date counts match. One query per model, selection, group and scope was independently sorted, giving 43,400 checks. Results for those 10,850 queries have repeated candidate controls; local scores for all 500,000 records do not.

Table S7: Matched Field/Subfield neighbour searches.

k	Field overlap	Subfield overlap	Difference (pp)	Lower / higher	Subfields
10	38.7%	36.9%	-1.73	145 / 72	217
25	46.7%	45.3%	-1.35	127 / 90	217
50	54.5%	53.7%	-0.79	109 / 108	217

Difference is Subfield minus Field. Each scope contains 256 candidates, including the same 50 query articles, with matched dates; self is excluded during search. Means use ten candidate selections. The broader Field comparison is conditioned on the queries and is not 256 unrestricted Field draws. All 217 Subfields appear in the paired design; 183 meet all size conditions of the separate 128/256/512 sensitivity panel. Full size/readout/k results accompany the table.

S4.2 Subfield persistence

The atlas compares ranks among the same 183 Subfields under 27 combinations of three candidate sizes, three readouts and three neighbour counts. Example rules were fixed after the overall results but before reading selected article titles. The six upper and six lower examples are descriptive extremes under those rules. Their rank ranges locate persistent relative agreement, not semantically correct or incorrect Subfields.

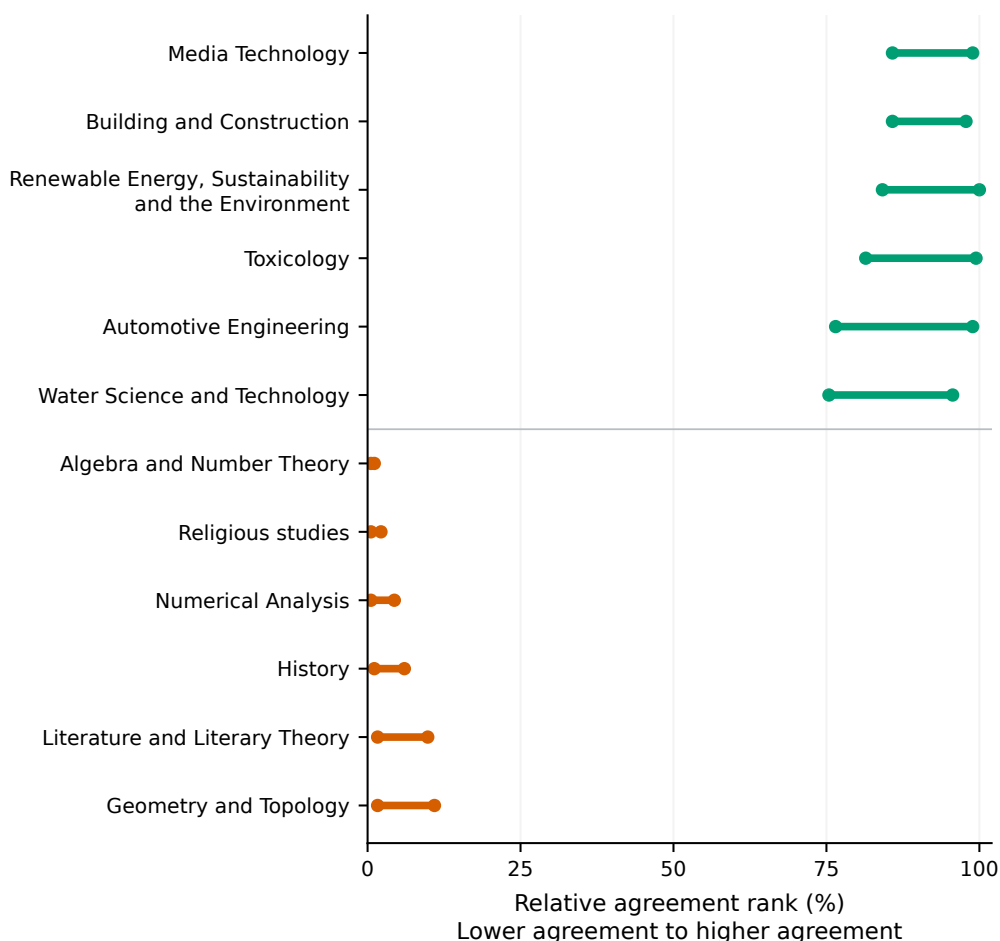


Figure S2: Relative neighbourhood-agreement ranks for twelve rule-selected Subfield examples. Ranges span 27 combinations of candidate size, readout and neighbour count, using the same 183 eligible Subfields. The upper/lower six illustrate the recorded higher/lower agreement rule. These are sensitivity ranges, not confidence intervals; higher agreement does not imply better thematic accuracy.

S4.3 Article and relation examples

The article atlas uses ten candidate selections for the fixed queries. Link comparisons are restricted to 531,650 directed relations whose endpoints remain eligible, avoiding the error of treating an unavailable target as a rejected neighbour. For each displayed source, both persistent and model-dependent targets are retained in the accompanying files. Figure S3 selects the eligible relation with maximal model disagreement for that source; it cannot estimate how common disagreement is.

The examples were inspected against saved OpenAlex records. The history-book, infection/politics and announcement cases remain visible, as does one target with an incomplete abstract. A diagnostic for 82 generic titles removes five fixed queries and barely changes

the overall query mean (45.344% to 45.349%). That narrow check neither estimates the prevalence of thematic classification errors nor cleans all candidates.

Table S8: All twelve rule-selected article examples, including source caveats.

Article and OpenAlex ID	Subfield label	Mean	Selection range	Inspection note
SIRT6 overexpression induces massive apoptosis in cancer cells but not in normal cells W2022976570; 2011	Geriatrics and Gerontology	79.3%	76.7%–82.4%	No clear issue in this limited inspection; not expert validation.
Comparison of The Effectiveness of Scientific Approach and Problem-Solving Approach in Problem-Based Learning in Class IX of SMP Negeri 3 Pangsid W2998141762; 2019	Applied Mathematics	84.1%	75.1%–87.4%	Mathematics education under Applied Mathematics; administrative label.
A systematic analysis on the managerial non-efficiency of the state-owned businesses W2359252036; 2002	General Decision Sciences	76.9%	74.7%–78.8%	No clear issue in this limited inspection; not expert validation.
Application of multi-agent approach to development of the petrol dispatching system of gas stations network W3107610255; 2020	Discrete Mathematics and Combinatorics	78.2%	74.0%–82.4%	No clear issue in this limited inspection; not expert validation.
Persuasive Talk, Delegation, and Announcements W2252942328; 2012	Safety Research	78.2%	72.1%–83.3%	Persuasion economics under Safety Research; label caveat.
Road risk-perception and pedestrian injuries among students at Ain Shams University, Cairo, Egypt W2081778888; 2011	Safety, Risk, Reliability and Quality	76.4%	70.8%–85.2%	No clear issue in this limited inspection; not expert validation.

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Table S8 (continued)

Article and OpenAlex ID	Subfield label	Mean	Selection range	Inspection note
Invasion Strategy W2011180849; 2003	Political Science and International Relations	18.7%	16.1%– 21.0%	Bacterial-infection abstract labelled as politics.
Beyond the witch trials: Witchcraft and magic in Enlightenment Europe W7045581866; 2004	Nuclear and High Energy Physics	21.4%	19.2%– 23.7%	History-book description labelled as physics; type caveat.
Optical properties of cobalt clusters implanted in thin silica layers W1966552042; 2006	Ceramics and Composites	23.2%	21.3%– 24.4%	Substantial formula code in abstract; input caveat.
Announcement W4234898602; 2010	Ophthalmology	22.2%	19.9%– 25.8%	Announcement / atlas review; not detected by the original notice screen.
Conversion of a rule based to an object oriented expert system W2117277818; 2002	Software	25.0%	22.8%– 26.0%	No clear issue in this limited inspection; not expert validation.
‘Thare is na leid on life of lordship hym like’: linguistic means of depicting king arthur in the poem Golagros and Gawain W3110667008; 2020	Classics	25.5%	24.4%– 26.3%	No clear issue in this limited inspection; not expert validation.

Numbers are k25 overlap averaged over 45 model pairs; ranges span ten candidate selections. The first/last six examples come from high/low agreement sources under the recorded rule. Titles were read after selection. Full source/target IDs, titles and per-model supports for both relation types are in the accompanying CSVs. The most model-dependent link was selected for each source; this does not estimate prevalence. An incomplete target abstract is also retained and marked in Figure S3. None of these checks validates OpenAlex labels.

SIRT6 / cancer [b]	10	9	10	10	10	10	10	0	1	10
Mathematics education [a]	4	7	4	0	8	5	7	0	7	10
State-owned businesses	9	2	3	10	10	8	0	10	10	10
Petrol dispatch	10	1	0	10	10	2	7	10	10	10
Persuasive talk [a]	4	10	10	10	4	8	4	10	0	10
Pedestrian injuries	10	8	3	10	10	1	0	10	10	9
Invasion Strategy [a]	0	10	10	0	10	10	10	0	3	0
Witch trials [a]	0	10	0	10	10	10	0	10	10	10
Cobalt / silica layers	10	10	10	0	0	10	10	0	0	10
Announcement [a]	0	0	8	10	10	0	0	10	0	10
Expert-system conversion	0	0	0	10	10	10	10	10	10	1
Arthurian poetry	0	0	10	10	10	9	10	10	10	10
	SPECTER	SPECTER2	SciNCL	SciBERT	BERT	MPNet	MiniLM	PubMedBERT	BioBERT	SimCSE

Figure S3: Most encoder-dependent eligible link for each of the twelve fixed article sources. Each cell counts selections in which the target is among the 25 nearest neighbours (0–10); both endpoints are always eligible within the 256-candidate design. Upper/lower six rows are high/low agreement sources. [a] Source or label caveat; [b] the selected target has an incomplete abstract. Full IDs, titles and caveats are retained in Table S8’s files. Links were selected for maximal model disagreement, so this panel does not estimate its prevalence.

S5 Time, discipline composition and model traits

These additional comparisons ask where agreement differs: across publication periods, areas and groups of models. Their controls assess possible explanations without identifying a single cause.

S5.1 Period contrasts and search opportunities

Period summaries give equal weight to Fields and fixed model pairs. Endpoint differences compare 2020–24 with 2000–04; the full five-period sequence determines whether the trajectory is monotone. Native candidate pools and 2,048-candidate pools are reported separately. A later identical-query comparison separates changes in selected queries from candidate count. Publications are represented by the same present-day models across all dates; differences may reflect coverage, writing, composition, text length or model exposure rather than historical changes in scientific organisation.

Table S9: Temporal endpoint changes and Field heterogeneity.

Measure / candidate design	Mean change	Positive	Negative	Non-monotone
CKA, native groups	0.0207	23	3	22
CKA, 2,048 per group	0.0206	24	2	22
Distance ranks, native	0.0304	22	4	19
Procrustes, native	0.0019	15	11	24
k25, all candidates	-0.0087	10	16	22
k25, 2,048 / mean	0.0091	20	6	20
k25, 2,048 / CLS	0.0034	16	10	22
k25, 2,048 / SEP	0.0138	21	5	20

End minus start: 2020–24 versus 2000–04. Each row describes 26 Fields with equal weight; neighbour changes are fractions, so multiply by 100 for percentage points. Five periods are used to classify monotonicity. Full k10/k25/k50, recipe, identical-query and pair-level checks are in the CSVs. Candidate counts can change the sign. These are descriptive period contrasts, not causal historical convergence.

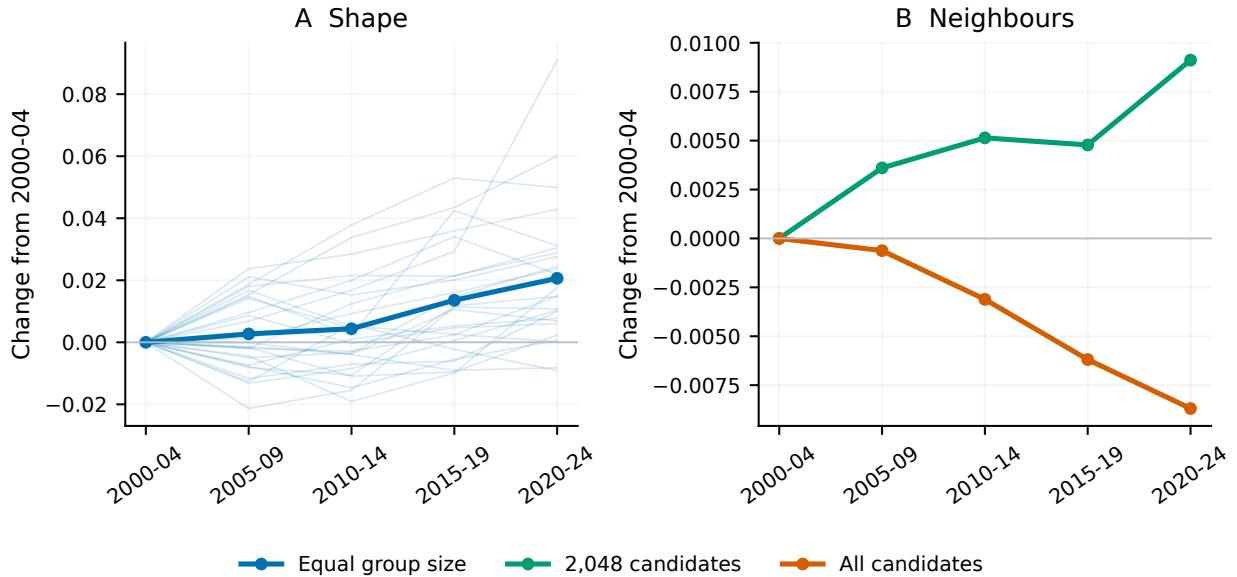


Figure S4: Descriptive time patterns under candidate controls. Values are changes from 2000–04. Fields and fixed model pairs have equal weight. In (A), thin lines show the 26 Fields; the thick line is their mean. In (B), k25 overlap is compared with all candidates or 2,048 candidates. Changes in search opportunity can reverse the average sign. These are not causal historical trajectories.

S5.2 Medicine and Subfield composition

Composition controls draw 2,048 articles per Field with identical date quotas under observed or balanced Subfield composition, repeated ten times. They are also summarised after excluding BioBERT and PubMedBERT. Energy has only one eligible Subfield in this control, so its composition does not change in the same way as Medicine’s. Balancing and model omission are descriptive sensitivities, not interventions isolating independent causes. The complete 26-Field results accompany the four displayed examples.

Table S10: Discipline agreement with observed and balanced Subfield composition.

Field	Models	CKA obs.	CKA bal.	k25 obs.	k25 bal.
Medicine	Ten	0.546	0.543	29.0%	29.2%
Medicine	Eight	0.564	0.562	29.5%	29.9%
Energy	Ten	0.772	0.772	33.2%	33.2%
Energy	Eight	0.774	0.774	33.4%	33.4%
Mathematics	Ten	0.671	0.644	29.4%	28.8%
Mathematics	Eight	0.691	0.675	30.8%	30.3%
Physics and Astronomy	Ten	0.663	0.674	33.0%	34.1%
Physics and Astronomy	Eight	0.665	0.671	33.3%	34.3%

Eight excludes BioBERT and PubMedBERT. 2,048 articles per Field with identical date quotas; mean of ten selections. Energy has one eligible Subfield, so balancing does not introduce a comparable composition change there. The display focuses on the four Fields of Figure S5; all 26 Fields and matched contrasts are supplied. Composition and biomedical training are not isolated causal explanations.

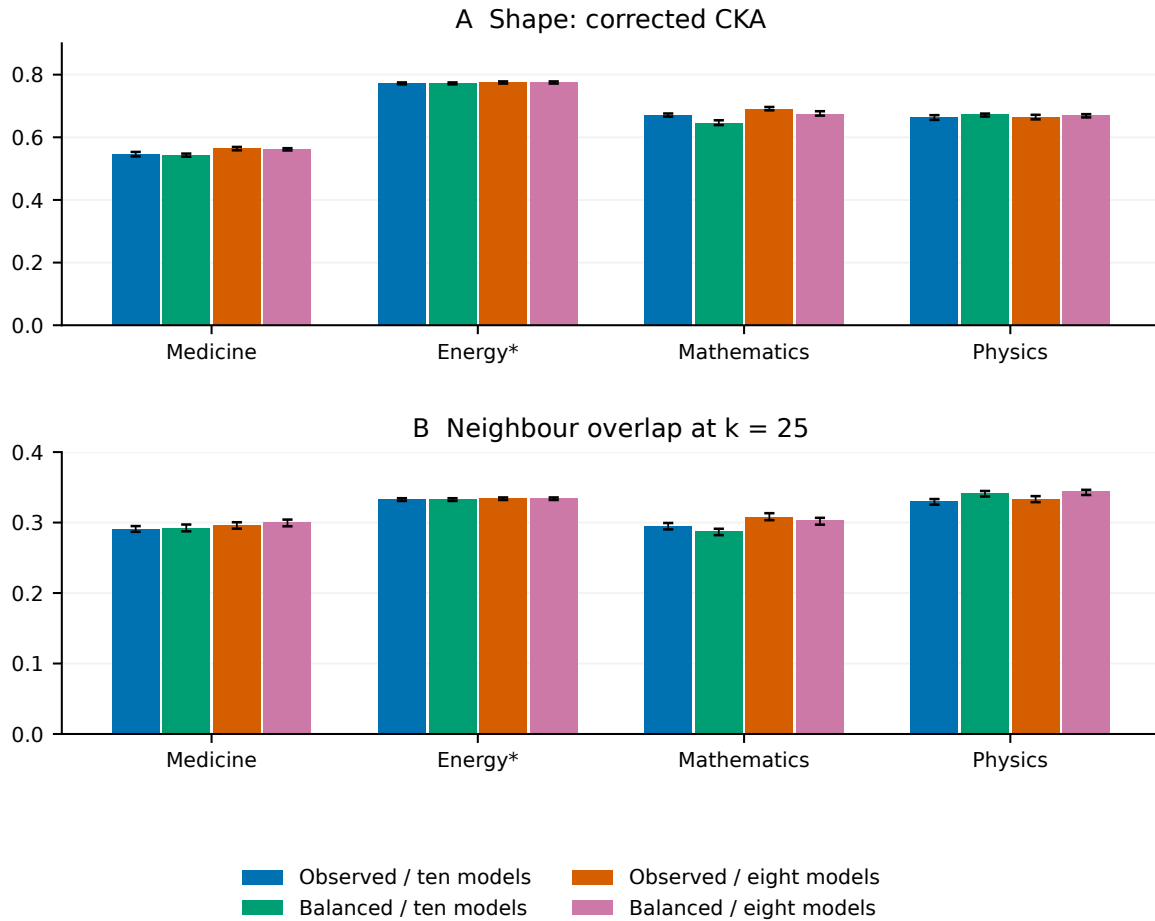


Figure S5: Observed and balanced Subfield composition in four Fields. Each condition uses 2,048 articles per Field and identical date quotas. Bars are means and whiskers the range of ten selections, not confidence intervals. The eight-model panel excludes BioBERT and PubMedBERT. *Energy has one eligible Subfield. Balancing composition and omitting biomedical models do not isolate a unique causal explanation. All 26 Fields are retained in Table S10's files.

S5.3 Readout and apparent model families

Recorded model traits cover broad training-domain and objective groups. Pairwise models, leave-one-model checks and readout scenarios assess their descriptive association with agreement. Shared model membership makes the 45 pairs dependent. Some traits are linked in this panel, and coefficients can become non-identifiable under omission; for example, the SEP scenario without SimCSE retains the missing coefficients rather than reporting spurious estimates. The heatmaps show how apparent families depend on the representation recipe.

Table S11: Descriptive model-trait fits and identifiability.

Outcome	Rank	Condition no.	R squared	Unidentified coefficients after omission
CKA / primary	7/7	7.91	0.741	0
CKA / 2,048 articles	7/7	7.91	0.741	0
CKA / quality filter	7/7	7.91	0.752	0
CKA / CLS	7/7	18.59	0.459	14
CKA / SEP	7/7	8.33	0.258	7
Neighbours k25 / mean	7/7	7.91	0.759	0
Neighbours k25 / CLS	7/7	18.59	0.507	14
Neighbours k25 / SEP	7/7	8.33	0.442	7

Ten fixed models yield 45 overlapping pairs. R squared describes these pairs, not predictive validity. The final column counts unavailable coefficient rows across the ten leave-one-model checks, not independent models or tests. In the SEP design, omitting SimCSE makes coefficients non-identifiable; those values remain missing. Architecture, lineage, final corpus, objective, domain and readout are correlated traits. Full trait definitions, source URLs, coefficients, pair-label permutation references and omissions accompany this table. No causal training effect is claimed.

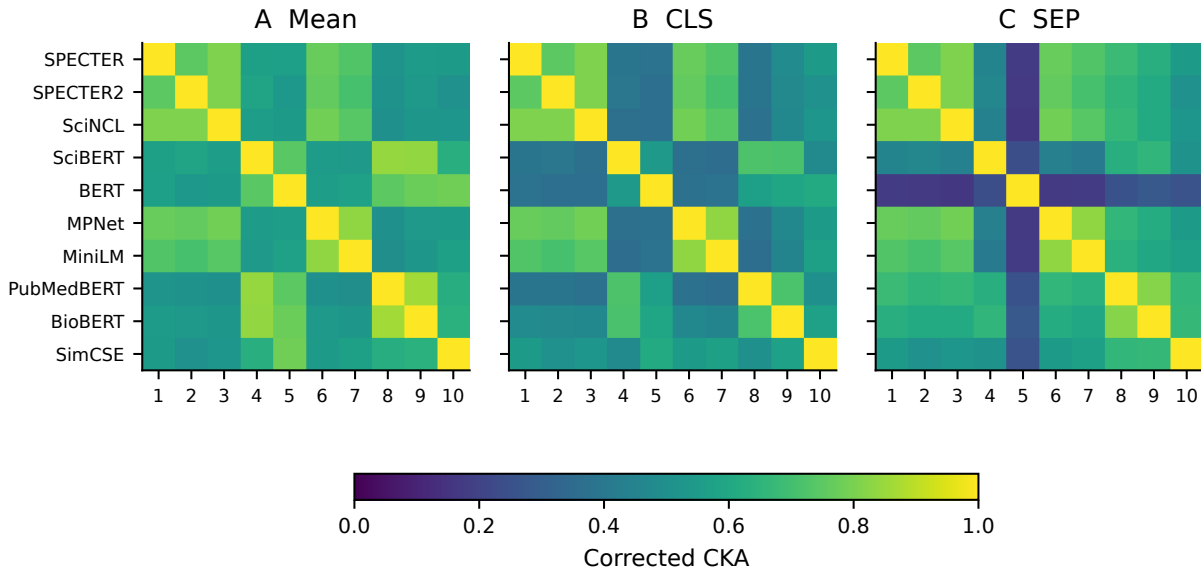


Figure S6: Pairwise shape agreement under three readout choices. Matrices show mean corrected CKA over the original 130 Field-period cells. Only BERT, SciBERT, BioBERT and PubMedBERT change readout; diagonal cells are identities. Model order in columns: 1 SPECTER, 2 SPECTER2, 3 SciNCL, 4 SciBERT, 5 BERT, 6 MPNet, 7 MiniLM, 8 PubMedBERT, 9 BioBERT, 10 SimCSE. Similarity patterns are not an isolated effect of shared training.

S6 Geometric properties and construct checks

These checks ask what a numerical property actually describes. A measure can repeat reliably and still fail to distinguish separate groups from an elongated cloud or a few extreme points.

S6.1 Properties, alternatives and selection stability

The geometric analysis reuses the 52,000-article input panel, with 2,000 articles per Field and equal period counts. All vectors have unit length and calculations use 64-bit arithmetic. For distinct vectors u_i, u_j , angular spread uses the median pairwise chord distance, converted to degrees as $2 \arcsin(q_{0.5}(\|u_i - u_j\|)/2)$. The saved implementation interpolates distance quantiles before this monotone conversion; the same convention is retained throughout. Alternatives are its 10th/90th percentiles and the median angle to the centre direction. They describe spread rather than semantic diversity or isotropy (Rudman et al., 2022).

After centring the unit-vector cloud, let λ_j be the covariance eigenvalues and $p_j = \lambda_j / \sum_l \lambda_l$. The participation ratio is $PR = 1 / \sum_j p_j^2 = (\text{tr } C)^2 / \text{tr}(C^2)$, where C is the centred covariance matrix. This is the main measure of effective linear dimension. The entropy alternative is

$\exp(-\sum_j p_j \log p_j)$, applied to the covariance spectrum (Roy and Vetterli, 2007). D80 counts the smallest number of descending eigenvalues containing 80% of variance. These quantities are unchanged by adding zero coordinates, so the 384- versus 768-coordinate comparison is not normalised by ambient dimension. They describe linear variation, not nonlinear intrinsic dimension or a count of topics.

Twenty half samples contain 1,000 articles per Field, and five further equal-size selections contain 2,000, all drawn from the fixed corpus with matched IDs across models. Central 90% ranges and absolute median shifts are divided by the primary value. Operational thresholds are 5% for angular spread, 10% for PR and 20% for connection. With five repeats, quantiles are coarse; individual values are retained. Passing those selections does not erase the four PR half-sample alerts. The complete analysis contains 16,884 sets of measures, including time, size and representation controls; not every alternative was repeated under every selection.

Table S12: Selection stability of the geometric properties.

Property	Selection design	Cases	Alerts	Median width	Maximum width
Angular spread	20 half samples	260	0	0.79%	1.81%
Angular spread	5 equal-size selections	260	0	0.49%	1.85%
Effective linear dimension	20 half samples	260	4	4.06%	12.84%
Effective linear dimension	5 equal-size selections	260	0	2.60%	7.53%
Connection	20 half samples	260	258	18.62%	423.07%
Connection	5 equal-size selections	260	51	10.59%	289.69%

A case is one model-Field combination. Width is the central 90% range divided by the primary value; the screen also checks median shift. Thresholds: 5% angular spread, 10% PR, 20% connection. These are operational sensitivity thresholds, not journal standards or population precision. The four PR half-sample alerts remain visible even though the five equal-size selections pass. PR is effective linear dimension, not number of topics. Spectral alternatives share the same covariance spectrum and do not independently validate thematic meaning.

S6.2 Sample size

This check varies the number of articles per area while keeping the model and measure fixed. Figure S7 shows how far each value is from the result at 2,000 articles per area.

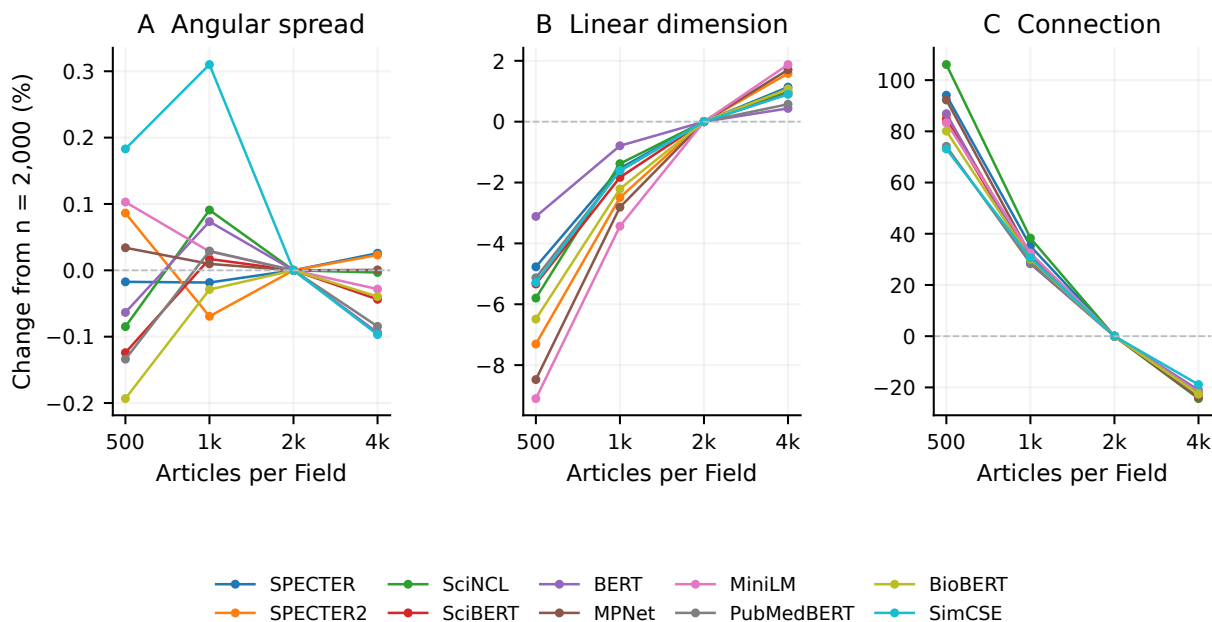


Figure S7: Sample-size sensitivity of the three saved geometric properties. Each line shows one model’s median relative change across the 26 Fields, compared with 2,000 articles per Field. Panels use different vertical scales. Connection uses the union-graph spectral gap with $k = 25$ fixed as n changes; the separate approximately fixed-fraction check is supplied in Table S13’s files. These curves do not certify population precision.

S6.3 Representation and content controls

Controls vary quality flags, title/abstract input, readout, common content, extreme points, global centring and the MiniLM token limit. The quality comparison replenishes each period to its specified size after applying recorded flags. The extreme-point control removes the ten points farthest from the centre in each period of the initial half sample; removing ten random points per period supplies its equal-size reference. It does not presume that extreme points are erroneous.

Global centring subtracts the mean of the balanced 52,000 unit vectors and then normalises each result again. This changes the representation and is distinct from centring a covariance matrix for PR. Three Gaussian clouds per model–Field case provide geometric references with matching mean/covariance only in expectation before unit normalisation. Their realised moments are checked; these references are neither exact matched-moment controls nor calibrated statistical null distributions. The point controls in Figure S8 and Table S16 have their own matched references and must not be treated as another complete set of repeated selections.

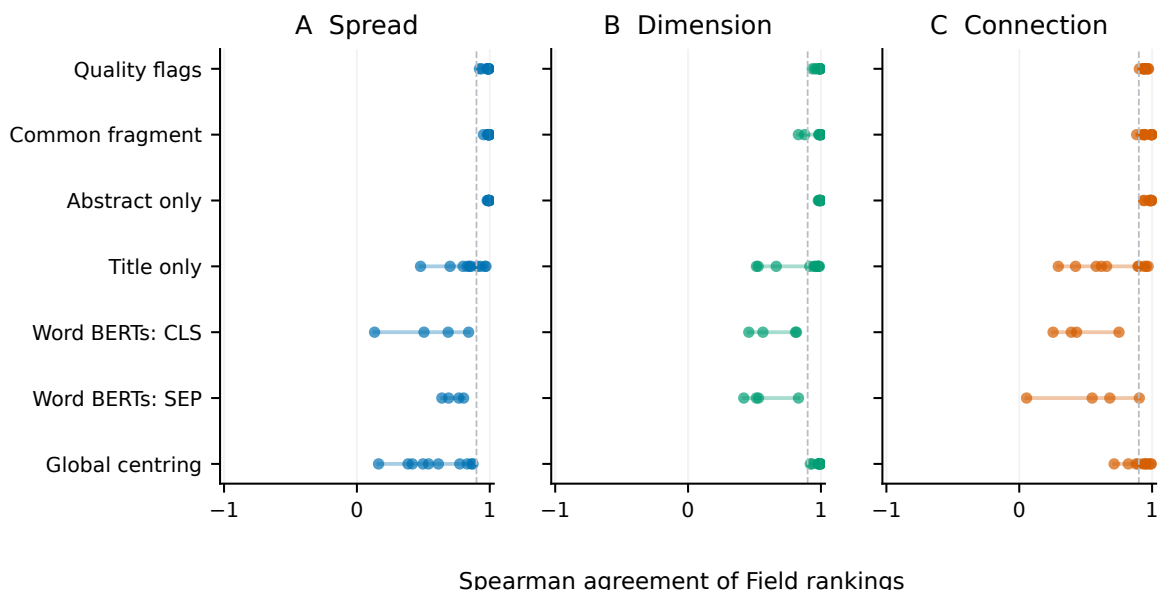


Figure S8: Agreement of Field rankings before and after each matched control. Each point is a model’s Spearman rank agreement; horizontal spans show their range. CLS/SEP change only the four word BERTs. The dashed line at 0.9 is a visual reference, not a pass/fail threshold. Global centring changes the representation. All saved controls, including additional point sensitivities, accompany Table S12.

S6.4 Why connection is not general fragmentation

Connection uses the second-smallest eigenvalue of the normalised graph Laplacian $L = I - D^{-1/2}AD^{-1/2}$ on the unweighted union of directed nearest-neighbour links. The value is zero for a disconnected graph. Primary $k = 25$ is checked at 10 and 50, with a weighted local-scale graph and mutual-neighbour components as alternatives. Spectral-clustering theory motivates the construction (von Luxburg, 2007); it does not establish that a small gap means separate scientific themes. Eigenvalue residuals and independent dense calculations were checked.

Single-linkage merging supplies the radii at which the largest component reaches specified fractions of the cloud. Radius ratios and the maximum-to-median minimum-spanning-tree edge ratio provide different connection descriptions. Nine synthetic clouds, followed by Gaussian references and article controls, exposed non-equivalent responses to elongation, isolated outliers and separated components. Fixed k also changes its fraction of candidates as sample size grows. The approximately fixed-fraction comparison was added during execution and is identified as a later sensitivity. These failures are why connection is excluded from the manuscript’s interpretation of Field fragmentation.

Table S13: Construct checks on all nine saved synthetic clouds.

Cloud	Gap k25	Radius 90/50	Max/median edge	Mutual giant
One round cloud	0.499	1.121	1.371	96.6%
One narrow cloud	0.503	1.132	1.400	95.5%
One wide cloud	0.487	1.084	1.262	99.8%
One elongated cloud	0.091	1.115	1.299	99.1%
One low-rank cloud	0.098	1.589	4.913	99.9%
Two separated groups	0.000	1.766	2.253	49.5%
Four separated groups	0.000	1.003	1.587	24.9%
Two groups with bridges	0.007	1.005	1.428	97.8%
One cloud with outliers	0.499	1.123	4.770	95.7%

Each saved cloud has 1,000 points. Smaller gap can arise without separate groups; radius/edge ratios can respond to outliers or elongated clouds. In all 260 primary model-Field cases the union k25 graph is connected. Connection therefore remains a graph description and is not validated as general thematic fragmentation. The five equal-size selections retain 51 connection alerts (Table S12).

The 780 Gaussian reference rows are included. Observed-relative reference errors span -4.69% to 4.60% for angular spread, and -10.47% to 19.78% for PR. Gaussian draws preserve fitted moments only in expectation before renormalisation; they are not calibrated null tests. Full n/k controls accompany the table; the fixed-neighbour-fraction check was added during execution.

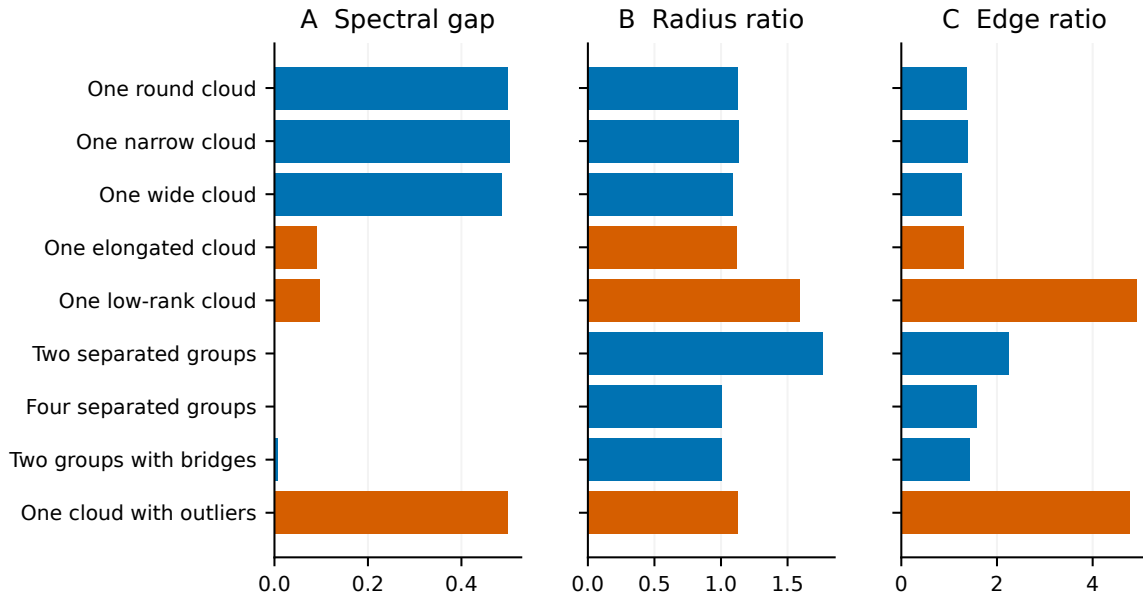


Figure S9: All nine saved synthetic-cloud checks. (A) Spectral gap of the union k25 graph; smaller means weaker connection. (B) Ratio of radii at which the largest component contains 90% and 50% of points. (C) Maximum divided by median minimum-spanning-tree edge length. Orange marks the elongated, low-rank and outlier clouds: apparent weak connection or late merging need not mean separate thematic groups. These are controlled geometric examples, not a calibrated statistical null.

S7 All Field-pair comparisons

Here, a pair is simply two different research areas, such as Medicine and Energy. The 325 pairs cover every such comparison among the 26 areas. The calculation asks which area has the larger geometric value, and whether changing the model reverses that answer.

S7.1 Directions and magnitude

All unordered Field pairs are enumerated by ascending OpenAlex ID. For each model and property, the 26 saved conditions are the primary selection, twenty half samples and five additional equal-size selections, with repeat indices matched across the two Fields. If $d_r = 2(B_r - A_r)/(A_r + B_r)$, direction is positive when $\min_r d_r > t + 10^{-10}$ and negative when $\max_r d_r < -t - 10^{-10}$. Otherwise it is unresolved. The minima $t \in \{0, 0.01, 0.05, 0.10\}$ were specified before the pair counts, after the morphology results. They are magnitude sensitivities, not tests of significance.

All-ten agreement requires every model to have the same persistent direction. Opposition requires at least one persistently positive and one persistently negative model. All remaining pairs are unresolved, including partial agreement. These classes are exhaustive and disjoint. The full range, rather than a favourable central interval, determines persistence; the overlapping selections do not constitute independent replications.

Table S14: All Field-pair conclusions under four minimum relative differences.

Property	Minimum	All ten agree	Opposition	Unresolved	Pairs
Angular spread	0%	42	262	21	325
Angular spread	1%	30	238	57	325
Angular spread	5%	10	96	219	325
Angular spread	10%	0	19	306	325
Linear dimension	0%	54	221	50	325
Linear dimension	1%	50	213	62	325
Linear dimension	5%	36	177	112	325
Linear dimension	10%	26	126	173	325

Every property uses all 325 Field pairs and ten models. A model direction must persist in all 26 saved conditions: primary sample, twenty halves and five equal-size selections. Opposition needs at least two fixed models with persistent opposite directions; unanimity needs all ten. Other cases are unresolved, not equal. The signed symmetric difference is $2(B - A)/(A + B)$; the minimum is an operational magnitude cutoff, not statistical significance or a share of science correctly represented. Full 650 pair-property rows and all 6,500 model directions accompany the table.

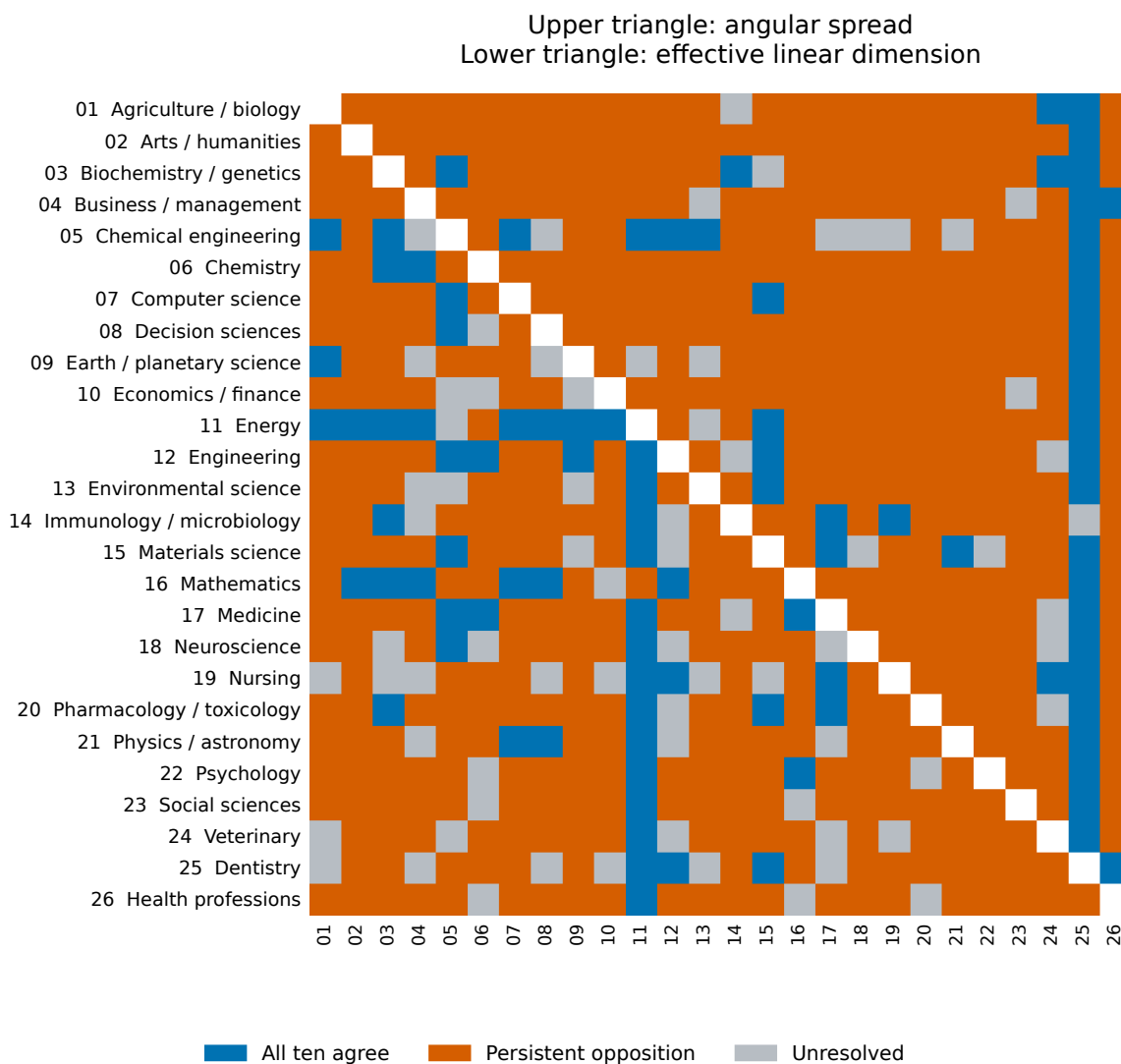


Figure S10: All 650 Field-pair/property classifications at zero minimum relative difference, using the 26 saved selection conditions. The upper triangle shows angular spread; the lower shows effective linear dimension (PR). Row labels and column numbers identify the same 26 Fields; names are shortened for display. Each triangle contains 325 pairs. Blue requires all ten models to agree persistently; orange requires at least two persistently opposing models; grey is unresolved. The diagonal is not a comparison. Full labels, model directions and magnitude checks are in Table S14's files.

S7.2 Alternative measures and model panels

An opposing model pair is retained under alternatives only when those same two models preserve their original directions in every alternative. Angular alternatives, entropy dimension and D80 now all cover the primary selection, twenty half samples and five additional selections: 26 conditions each. The original three-condition dimension check is preserved as history; Table S15 uses the completed coverage. D80 is discrete and ties remain unresolved. At positive cutoffs, PR must exceed the stated minimum; its alternatives must preserve nonzero direction. Models are not replaced with different witnesses to preserve a count. Ten leave-one-model panels and the six similarity-oriented models test panel dependence; smaller panels offer fewer opportunities for opposition.

Table S15: Persistent opposition with full alternative-measure coverage.

Model panel	Models	Spread	Spread + alternatives	PR	PR + alternatives
All ten	10	262	225	221	190
Six similarity models	6	231	190	160	115
Omit SPECTER	9	234	182	212	174
Omit SPECTER2	9	251	205	211	173
Omit SciNCL	9	259	217	215	182
Omit SciBERT	9	258	222	218	187
Omit BERT	9	261	223	211	176
Omit MPNet	9	262	223	218	186
Omit MiniLM	9	259	223	221	188
Omit PubMedBERT	9	261	220	206	187
Omit BioBERT	9	259	217	212	169
Omit SimCSE	9	259	217	217	183

Counts are out of 325 at zero minimum magnitude. Every angular and dimension alternative now covers all 26 selections. The same opposing models must retain their directions; substitutes do not count. D80 ties remain unresolved. The six-model panel comprises SPECTER, SPECTER2, SciNCL, MPNet, MiniLM and SimCSE. Full cutoffs and directions are supplied as CSVs.

S7.3 Matched point controls

Each of the eleven point controls is compared with its appropriate equal-size reference. Changing quality, extremes or sample size changes IDs by design; other matched controls verify identical IDs. The summary distinguishes a changed class from survival of the same original opposing models and directions in both reference and variant. One favourable con-

trolled comparison does not demonstrate persistence across 26 new selections. The saved differences were independently audited through equivalent ratio inequalities, covering 442,000 contrasts without importing the classification function.

Table S16: Point controls: retention of the original opposing models.

Control	Matched reference	Models changed	Spread retained	PR retained
Title only	half	10	246/262	193/221
Abstract only	half	10	262/262	221/221
Word BERTs: CLS	half	4	253/262	210/221
Word BERTs: SEP	half	4	257/262	201/221
Common fragment	common_native	10	262/262	216/221
Global centring	half	10	145/262	218/221
Quality flags	half	10	262/262	221/221
Extreme-point removal	trim_random	10	262/262	221/221
MiniLM 512	primary	1	262/262	221/221
n = 500	primary	10	262/262	219/221
n = 4,000	primary	10	262/262	221/221

Numerators retain the same original model pair with its original signs in both matched reference and variant. Denominators are the original persistent oppositions (262 angular spread; 221 PR). Values are not new 26-condition stability tests. The common fragment uses its own matched panel; extreme-point removal compares with random removal at equal size. Quality/extreme/size controls change IDs as specified; other matched controls retain IDs. All magnitudes and all 7,150 point pair records are included.

S8 Reproducibility and access

Table S17: Public release archives and their SHA-256 checksums.

Contents	Archive / SHA-256	Size
Code and guides	code-and-guides.zip dbc3a4b4e475866244d698abfc5f003920389b4c1ead9c6e9 ed88303bbc2f104	24.15 MB
Corpus metadata	data-analysis-ready-v1.zip b33ea5276587db344878cd3afc5c28c8f39274894d09e9c50 e26397225a3be65	44.66 MB
Initial comparisons	data-analysis-v1.zip bded3d05d03f2c2556f6d68246b99e9d5e11f9e7a614d3ef2 a55263fce2f471e	75.49 MB
Text pilot	data-checklist-v1.zip 0615859291770cef75bdc38cf4f367c516a86bd0a7d193eca 71f658cbb0739c1	14.71 MB
Model configuration	data-embeddings-v1.zip cf2a01517e1e30178f4469017eadc259c36ab59d692d18e51 ef92e8f5a7e6c9f	0.04 MB
Area pairs	data-field-pair-summary-v1.zip 71d7ddb30bed949dd01e15c8611740c10c72dcb815c8ea830 9d6fd80a24afda1	3.35 MB
Geometry	data-morphology-pilot-v1.zip 372b07a29f77a0eb4d9ac8d34893669e7eee04efb06adc4a3 cf585151d9a0836	11.00 MB
Source cases	data-prepaper-v1.zip e10a813171cd3992bebd9cb5f2bd92f7e8e630a9b63d13962 64ff23f05038ade	2.54 MB
Final controls	data-robustness-closure-v1.zip 9454a5f54f63c4dbbe16d0888778d70128d12257b34c2fb53 b5c047e4c6f287b	106.99 MB
Expanded comparisons	data-robustness-v2.zip 7b34526a5d4097fa9e74bc39d7dc5ebbf40d313485d3c2a4 e1e245b3717842e	209.20 MB

All listed archives are public under DOI 10.5281/zenodo.22876602. `SELECTION.json` lists the selected data files and their origins. These are analysis-level data; historical texts and vector caches are excluded. Code is MIT, original results/documents CC BY 4.0, and OpenAlex metadata CC0.

S8.1 Rebuilding this display package

The repository command `./manuscript/build.sh` compiles the included LaTeX into the main document and this supplement. It does not run models, searches or simulations. The compiler may download standard TeX support files on its first use. Re-exporting plots and table summaries is a separate presentation command, documented in the package README.

S8.2 Public statistical reproduction

The compact data and code release is archived at <https://doi.org/10.5281/zenodo.22876602> (Treny Ortega, 2026). After extracting the code and data archives together, `python reproducibility/reproduce_analysis.py` verifies the supplied hashes and recalculates 25 tables with 106,445 rows. It checks 17,550 local neighbour scores from per-query intersection counts, recalculates the balanced local and global averages, and runs the aggregation checks for all four main figures. The regenerated table contents match the saved references exactly both locally and in a fresh Linux environment with the pinned dependencies. Pinned dependencies, commands and tolerances are provided in the release README.

This route starts from frozen model-derived measurements. It does not regenerate embeddings from historical text, reconstruct the excluded nearest-neighbour lists or certify a new run of every supplementary analysis. Historical title/abstract input tables are withheld because redistribution rights to the plaintext abstracts have not been established. Exact identifiers and input hashes are retained, but fresh OpenAlex retrieval may differ. Manuscripts and supplements are separate from the data release and may be edited without changing the archived measurements.

S9 Final targeted robustness checks

These checks were specified after the previous results and the initial manuscript, before their new outcomes were inspected. They reuse the frozen vectors, preserve all earlier outputs and address the remaining selection, processing and source-quality questions. They add no model training or corpus extraction. Tables S18–S20 are accompanied by unrounded CSV summaries; the repository’s `data/robustness_closure_v1/` holds selections, source snapshots and per-part checksums. `ROBUSTNESS_CLOSURE_PROTOCOL.md` records the design. The separate data release supports the statistical reproduction described in S8.

S9.1 Repeated centres and direct neighbour contrasts

The 256-article centre check uses fifty selections of the same 26 Fields and 217 Subfields. Within each group, selection preserves the original five period counts. The size check fixes 26 Fields and 183 Subfields at 128, 256 and 512 articles; each has fifty repetitions. Period quotas derive from the original 512-article prefix, with nested largest-remainder rounding for smaller sizes. Thus its 256-article quotas differ from the first check. One within-period reassignment accompanies every selection, using exactly the same articles across observed/random groups and models.

The restricted comparison selects one eligible Subfield per Field fifty times, then selects articles ten times within each choice. Between-choice standard deviation of the observed mean is 0.01205; mean within-choice standard deviation is 0.00157. These are descriptive components, not independent population variances. The old restricted random reference drew 26 centres from a reassignment across all 217 Subfields; the new reference instead shuffles exactly the selected 26 groups’ articles. Omitting each Field from the original centres changes mean CKA by at most 0.00687, for Mathematics; the largest individual model-pair shift is 0.02355 for MiniLM–BioBERT. All omissions are also calculated across the fifty new selections.

For direct neighbour stability, each of fifty selections draws 200 papers per Field and period from the text experiment: 1,000 candidates per Field. Queries and candidates coincide; self matches are excluded. Full exact candidate rankings can be restricted to these subsets without changing the search. Independent direct sorting checks 4,680 queries, in addition to synthetic checks with tied vectors. The contrast averages the ten same-model title changes and the 45 between-model title-plus-abstract changes, weighting Fields equally. A model-specific reference averages its nine model comparisons. Main k values are 10, 25 and 50; CLS/SEP affect only the four word BERTs and are checked at $k25$. Model-specific mean contrasts range from -14.04 to $+16.59$ percentage points; Field means range from -4.43 to $+5.63$. The grand mean is not a universal ordering or an equivalence test.

Table S18: Repeated centres and direct input–model neighbour contrasts.**A. Centres**

Groups	Papers	Mean	Selection range	Random mean
26 areas	256	0.940	0.937–0.943	0.627
217 specialties	256	0.911	0.910–0.912	0.636
26 specialties	256	0.904	0.878–0.926	0.660
Areas: size check	128	0.938	0.934–0.942	0.618
Areas: size check	256	0.940	0.937–0.943	0.626
Areas: size check	512	0.941	0.939–0.943	0.647
183 specialties	128	0.913	0.911–0.914	0.617
183 specialties	256	0.914	0.913–0.915	0.629
183 specialties	512	0.915	0.914–0.915	0.661

B. Neighbour replacement

Readout	k	Model change (%)	Title only (%)	Difference (pp)	Range (pp)
Main	10	70.20	72.17	1.97	1.85 to 2.08
Main	25	64.93	66.49	1.56	1.44 to 1.69
Main	50	60.11	61.15	1.04	0.94 to 1.18
CLS	25	72.82	69.70	-3.13	-3.22 to -3.03
SEP	25	69.15	67.15	-2.00	-2.10 to -1.88

Ranges are 2.5th–97.5th selection percentiles, not confidence intervals. A: 50 selections per row, except 26 specialties (50 specialty choices, ten article selections each). Size checks have their own fixed period quotas. Random groups use the same articles and dates. B: 50 selections, 1,000 candidates per area; difference = title-only replacement minus model-change replacement. No grand-mean sign reversals occur within a readout. CLS/SEP change only four word BERTs.

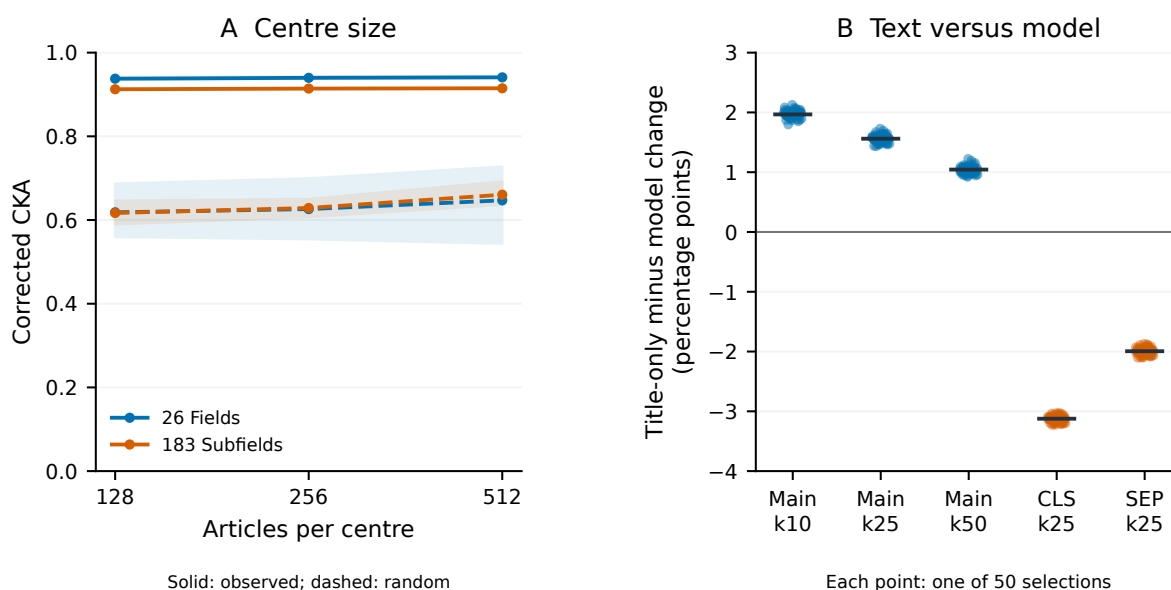


Figure S11: Additional selection checks. (A) Centre agreement at three article counts; solid lines show observed groups and dashed lines show matched random groups. Shading spans 2.5th–97.5th percentiles of fifty selection means. The same 183 Subfields are used throughout. (B) Each point is one of fifty balanced selections in the direct neighbour test. Positive values mean that removing the abstract changes more neighbours than changing models; negative values mean the reverse. Black marks are medians. Main uses the primary recipes; CLS/SEP change only the four word BERTs. Neither panel supplies population confidence intervals.

S9.2 Quality filtering with matched queries

The existing strict filter removes seven flagged categories and retains the lowest row index in each connected component of identical nonempty DOI or normalised text. This is not a new near-duplicate detector. It retains 448,886 records. Full searches compare every surviving query against original and filtered candidates in its Field-period group. At k25, the original all-query mean is 30.29%; restricting queries alone gives 30.58%, and filtering their candidates gives 31.43%. The matched-query gain is 0.85 points, or 0.77 after the existing chance adjustment.

The equal-count controls include the same 100 clean queries in both candidate pools, filling remaining places by deterministic random order. All 130 cells support 1,024 candidates; only 63 support 2,048. The largest Field-average change in full searches is 2.72 points in Mathematics. Across model pairs, the largest mean gain is 1.43 points for SPECTER2–SciBERT. The 52,000-paper text panel has at least 235 clean papers per Field-period cell. It supports 200 candidates and twenty clean queries per period in both arms, or 1,000 candidates and 100 queries per Field. Thus candidate counts, dates and queries stay matched.

All k values retain substantial model disagreement. The filter’s limited coverage prevents concluding that all source errors are harmless.

Table S19: Quality sensitivity with the same starting papers.

Candidates	Groups	k	Original (%)	Filtered (%)	Change (pp)
All	130	10	26.41	27.07	0.66
All	130	25	30.58	31.43	0.85
All	130	50	34.24	35.28	1.04
1024	130	10	30.71	30.91	0.20
1024	130	25	36.01	36.29	0.28
1024	130	50	40.80	41.11	0.31
2048	63	10	27.68	27.83	0.16
2048	63	25	32.06	32.25	0.19
2048	63	50	35.90	36.13	0.24

Matched input check: neighbours replaced

k	Model: original	Model: filtered	Title: original	Title: filtered
10	69.71	69.56	72.11	72.04
25	64.57	64.27	66.49	66.45
50	59.83	59.47	61.23	61.17

All values except changes are percentages. The filter retains 448,886 articles. The first block holds queries identical before and after filtering; it is distinct from the original all-query mean of 30.29% at $k=25$. Fixed-size searches use 100 clean queries included in both pools. Only 63 groups retain 2,048 candidates; all 130 support 1,024. The input check uses 100 queries and 1,000 date-balanced candidates per area, drawn from the saved text panel. It does not test every possible source error or near duplicate.

S9.3 Complete processing and dimension checks

For each model, the global mean is calculated once from its 52,000 unit vectors. Every original unit vector in each saved selection has that mean subtracted, then is normalised again. The mean is not recalculated per Field or repetition. Covariance centring alone would leave PR unchanged; the subsequent row normalisation makes this a different representation. Spread and PR use the original main selection, twenty halves and five further equal-size selections. Entropy dimension and D80 are completed on those same 26 original selections. No selection or magnitude cutoff is changed.

The classes and fixed-model direction rules remain those of S7. At zero cutoff, 142 original angular oppositions keep the opposition class after centring, but only 128 keep an original model pair with both directions unchanged; nine new oppositions give 151 in total. PR retains 204 original oppositions and adds seven, giving 211. The same-direction requirement also retains only five of the original 54 all-ten PR agreements jointly under entropy and D80; many comparisons cannot satisfy a rule requiring every model to retain a nonzero ordering. D80 ties stay unresolved. All class transitions, reversed witness directions and magnitude checks are included, not just favourable retentions.

Table S20: Full persistence after centring and under dimension alternatives.

Property	Cutoff	Original	Centred	Same models and answers	New	Now unresolved
Spread	0%	262	151	128	9	53
Spread	1%	238	113	78	12	112
Spread	5%	96	11	1	6	91
Spread	10%	19	1	0	1	19
PR	0%	221	211	204	7	16
PR	1%	213	201	194	6	17
PR	5%	177	156	151	4	24
PR	10%	126	106	100	4	23

PR oppositions retained by the alternatives

PR cutoff	Original PR	Entropy	D80	Both
0%	221	209	190	190
1%	213	202	180	180
5%	177	173	151	151
10%	126	126	114	114

Every condition uses all 26 selections and 325 area pairs. First block: original/centred are total opposition counts; same models and answers requires an original fixed witness pair to keep both directions. New oppositions were not original oppositions. Now unresolved counts original oppositions that become unresolved. At zero cutoff, 142 angular pairs and 204 PR pairs keep the opposition class; same-witness counts are 128 and 204. Reversing both directions occurs for 19 angular witness pairs in ten area pairs, and none for PR; these counts can overlap other categories. Second block: the cutoff applies to PR, while alternatives must preserve nonzero direction in every selection. D80 ties stay unresolved. Full class transitions and witness identities are in the CSV files.

The numerical audit independently reproduces 78,000 model-specific directions using equivalent ratio inequalities, reproduces the saved original neighbour means to numerical precision,

and verifies all 900 centre selections and matched references. None of these checks establishes thematic accuracy, independence of overlapping selections or population precision. The main text therefore distinguishes stable averages from 2,628 local selection alerts among 8,235 Subfield–model-pair comparisons.

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